

MANSI PATIL

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PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, CGPA 8.74/10) with hands-on experience across data analytics, machine learning, and Generative AI, built through an industry internship and independent end-to-end projects. Comfortable executing well-defined analytical tasks — data cleaning, feature engineering, exploratory analysis, and structured model validation — using SQL, Python, PySpark, and Power BI/Streamlit. Applied classification, clustering, and forecasting techniques (0.84+ ROC-AUC) and tracked experiments in MLflow for reproducibility. Also gained exposure to unstructured data and LLM-based techniques through applied RAG pipelines, cutting retrieved context by over 60% via relevance filtering and reranking. Seeking an entry-level Data Scientist role to support senior team members in building reliable models, dashboards, and AI-enabled solutions within structured, guided workflows.

TECHNICAL SKILLS

BI, Analytics & Data: Power BI, SQL, Python, PySpark, Pandas, NumPy, Feature Engineering, Data Cleansing & Validation, Exploratory Data Analysis, Matplotlib/Seaborn

Machine Learning: Scikit-learn, XGBoost, PyTorch, Regression, Classification, Clustering, Forecasting, Model Evaluation & Cross-Validation, OpenCV, YOLO

Generative AI / LLM: LLM Integration, Prompt Engineering, RAG, Multimodal RAG, Embeddings, Vector Databases (Qdrant), Cross-Modal Retrieval, LangChain, LangGraph, OpenAI GPT, Claude, Gemini, Groq API, Hugging Face

Backend, APIs & Databases: FastAPI, Django, REST APIs, MySQL, PostgreSQL, SQLite

Tools & Workflow: Docker, Git/GitHub, GitHub Actions, MLflow (experiment tracking), Jupyter, Streamlit, Postman

EXPERIENCE

Data Science Intern | *Amdox Technologies* ([GitHub](#))

Apr 2026 – Jun 2026

- Supported senior team members on **NeuralRetail**, an enterprise retail analytics system, by developing 3 ML models under guided workflows — **XGBoost churn classification (0.84+ ROC-AUC)**, ARIMA 30-day demand forecasting, and K-Means customer segmentation.
- Built and tested secure **RESTful FastAPI endpoints** with JWT authentication and SQLite persistence, powering real-time model scoring and automated inventory reorder alerts.
- Performed structured exploratory data analysis and data validation, and built an interactive **Streamlit/Plotly dashboard** to visualize sales trends and churn simulations.
- Tracked model lifecycle metrics for all 3 models using **MLflow**, following defined evaluation practices to support reproducible experimentation.

PROJECTS

Multimodal RAG Engine — Cross-Modal Document Intelligence System ([GitHub](#)) — *Python, FastAPI, Qdrant, CLIP, BLIP-2, Claude API, Docker*

- Built a multimodal RAG pipeline extracting text, tables, and high-DPI page images from PDFs (PyMuPDF), with sentence-aware chunking (~300 characters) to eliminate mid-sentence breaks.
- Designed a dual-vector retrieval system combining 384-dimension SentenceTransformer text embeddings and 512-dimension OpenCLIP image embeddings via Reciprocal Rank Fusion, enabling joint retrieval of text and visual chart content.
- Reduced retrieved context by over 60% (to 3 high-precision chunks per query) by applying a dynamic relevance cutoff (≥ 0.40) and MS-MARCO Cross-Encoder reranking in Qdrant, cutting noise while preserving answer accuracy.
- Engineered a grounded Vision-LLM generator (Claude API) with XML-based prompt-injection defenses, inline citations, and refusal logic to prevent hallucinated answers, validated by an automated regression-testing suite; deployed via FastAPI with a Streamlit UI.

Newsly — AI-Powered Daily News Digest Agent ([GitHub](#) | [Live Demo](#)) — *Python, FastAPI, Groq API (Llama 3.3-70B), Docker, GitHub Actions*

- Built an LLM-powered news agent retrieving and summarizing articles across 5 categories in real time using Google News RSS retrieval and prompt-engineered summarization with Llama 3.3-70B, in an applied RAG-style pipeline.
- Automated a FastAPI dashboard with scheduled Gmail SMTP digest delivery; containerized with Docker and orchestrated daily runs via GitHub Actions with zero manual intervention.

Loan Approval Prediction System with AI Chatbot ([GitHub](#) | [Live Demo](#)) — *Python, Scikit-learn, XGBoost, FastAPI, Streamlit, OpenAI API (GPT-4o-mini)*

- Delivered an end-to-end ML classification pipeline (XGBoost, Scikit-learn) to assess applicant financial risk and predict loan approval status based on 10+ demographic and credit variables.
- Integrated an AI financial advisory chatbot (OpenAI GPT-4o-mini) to explain prediction rationale and provide actionable, data-grounded, and interpretable recommendations.
- Deployed an interactive Streamlit analytics dashboard alongside the full-stack system, taking it live to cloud production via Render.

EDUCATION

Maharashtra Institute of Technology — B.Tech, Computer Science Engineering | **CGPA: 8.74 / 10**

CERTIFICATIONS

Business Intelligence with Advanced Excel and Power BI (2024) • CSRBOX — Agentic AI: From Learner to Builder – AI Agent Architect (2025) • Deloitte — Data Analyst Certificate (2025)